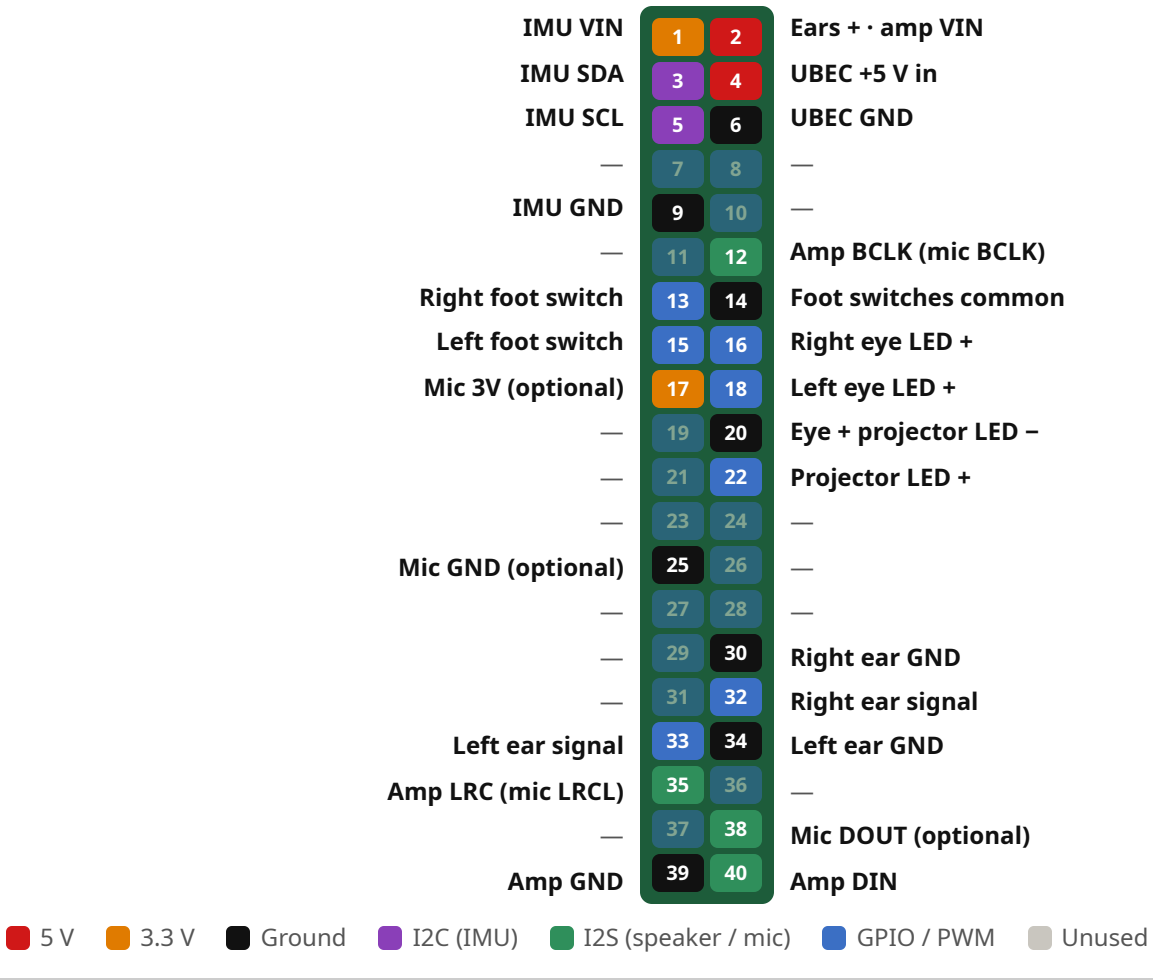
Open Duck Mini v2: pinouts

Every connection to the Raspberry Pi Zero 2 W, the servo bus and the power system, for this build.

> **Left and right are the duck's own sides.** Facing the duck, its left is on your right. Pin numbers are **physical header pins** (1–40), not GPIO numbers. Pin 1 is the square pad, at the end nearest the microSD slot.

Raspberry Pi 40-pin header



All ground pins (6, 9, 14, 20, 25, 30, 34, 39) are connected together on the board, so any of them works. Grounds are spread out here so no pin gets more than one or two wires. Both 5 V pins (2, 4) are the same rail, fed by the UBEC.

By device

Power in: UBEC (5 V for the Pi)

UBEC output	Pi pin	
+5 V (red)	**4**	5 V
GND (black)	**6**	GND

Set the UBEC jumper to **5.0 V**. Its input comes from the battery after the switch. The Pi is powered through these header pins, so the Pi's **PWR IN** micro-USB is left empty.

IMU: BNO055

IMU pin	Pi pin	
VIN	**1**	3.3 V
GND	**9**	GND
SDA	**3**	GPIO 2 (I2C SDA)
SCL	**5**	GPIO 3 (I2C SCL)
ADR	**—**	10 kΩ to GND on the IMU board → address **0x28**
3VO, RST, INT, PS0/1	**—**	not connected

The I2C bus runs at 50 kHz (`dtparam=i2c_arm_baudrate=50000` in `/boot/firmware/config.txt`) because of the Pi's clock-stretching bug with this chip.

Foot switches

Switch	Pi pin	
Left foot	**15**	GPIO 22
Right foot	**13**	GPIO 27
Common (both switches)	**14**	GND

The switch connects the pin to GND when the foot touches down. The Pi's internal pull-up holds it high otherwise, so no resistor is needed. Each foot has two switches: wire them **in parallel** (either one pressed counts as contact). Use the switch's **COM** and **NO** (normally open) tabs.

Ear servos (9 g micro servos)

Servo wire	Left ear	Right ear
🟠 Signal (orange / yellow)	**33 GPIO 13**	**32 GPIO 12**
🔴 +5 V (red)	**2**	**2 (shared)**
🟤 GND (brown / black)	**34**	**30**

The signal pins must be GPIO 12 and 13: they are the Pi's hardware PWM outputs, which drive the ears without jitter. Splice the two red wires into one lead for pin 2, or use pin 4 alongside the UBEC.

> **The official assembly guide's table swaps left and right for the ears and the eyes.** This sheet follows the code that actually runs on the duck. Both the official runtime and the version on your duck use: left ear = GPIO 13, right ear = GPIO 12, left eye = GPIO 24, right eye = GPIO 23. If an ear moves the wrong way once it's running, swap its two signal wires.

Eye LEDs and projector (optional)

LED	Pi pin	
Left eye + (anode)	**18**	GPIO 24
Right eye + (anode)	**16**	GPIO 23
Projector + (anode)	**22**	GPIO 25
All three − (cathodes, common)	**20**	GND

A GPIO pin gives 3.3 V and only about 16 mA. Put a **series resistor** (about 100 Ω) in each LED's lead unless the LED has one built in. Without it, the LED can damage the pin.

Speaker amplifier: MAX98357A (optional)

Amp pin	Pi pin	
VIN	**2**	5 V (shared)
GND	**39**	GND
BCLK	**12**	GPIO 18 (I2S clock)
LRC	**35**	GPIO 19 (I2S word select)
DIN	**40**	GPIO 21 (I2S data out)
GAIN, SD	**—**	not connected

The speaker connects to the amp's + and − screw terminal, not to the Pi.

I2S microphone (optional)

Mic pin	Pi pin	
3V	**17**	3.3 V
GND	**25**	GND
BCLK	**12**	GPIO 18 (shared with the amp)
LRCL / WS	**35**	GPIO 19 (shared with the amp)
DOUT	**38**	GPIO 20 (I2S data in)
SEL	**—**	to GND (left channel)

The official guide doesn't give microphone pins. This is the standard Pi wiring for an I2S MEMS mic, which shares the clock lines with the amplifier.

Camera (optional)

Ribbon cable into the Pi's **CSI camera connector**. A Pi Zero needs the narrow 22-pin Zero cable. Contacts face the board on the Pi end.

Servo adapter USB

Adapter USB-C → the Pi's micro-USB port marked **USB** (the one nearer the middle, not PWR IN). Plug it in **before** powering on, or it may not show up as `/dev/ttyACM0`.

Servo bus: Waveshare Bus Servo Adapter (A)

Adapter connection	Goes to
Screw terminal +	Battery + (XT30, after the switch), 22 AWG with a ferrule
Screw terminal −	Battery −, 22 AWG with a ferrule
Barrel jack	Not used (no reverse-polarity protection on that input)
USB-C	Pi micro-USB **USB** port
Jumper	Position **B** (USB control)
Servo connectors (×2)	Servo chains. Both are on the same bus, so it doesn't matter which chain goes in which socket.

The screw terminal input has reverse-polarity protection; the barrel jack doesn't. The servo sockets carry battery voltage directly (rated 6 A).

Joint	Right	Left
Hip yaw	**10**	**20**
Hip roll	**11**	**21**
Hip pitch	**12**	**22**
Knee	**13**	**23**
Ankle	**14**	**24**

Neck and head	ID
Neck pitch	**30**
Head pitch	**31**
Head yaw (±20° soft limit planned, harness)	**32**
Head roll	**33**

Servo IDs set by `setup_all_servos.py`, matching the runtime's joint names.

Power

From	To	Notes
2S 18650 pack → BMS	BMS pads 0V (B−), 4.2V (midpoint), 8.4V (B+)	Cells in the holder; bridge = midpoint
BMS − (P−)	14 A toggle switch → XT30 −	Load and charger negative go to **−**, never to 0V
BMS + (P+)	XT30 +	22 AWG main leads
USB-C charger	BMS + and −, on the battery side of the switch	Charges with the switch off
XT30 (robot side)	Servo adapter screw terminals, and UBEC input	Full pack voltage, 6.0–8.4 V
UBEC output (5.0 V)	Pi pins 4 (+) and 6 (−)	Powers the Pi and everything on its header

Full details and checklists: battery pack wiring sheet.

Software switches (`~/duck_config.json`)

Setting	Set it to
`imu_upside_down`	`true` if the IMU chip's +X points right, +Y back, +Z down. `false` if +X left, +Y back, +Z up. Check with `imu_axes_check.py`.
`expression_features.antennas`	`true` once the ear servos are wired
`expression_features.eyes` / `projector` / `speaker` / `microphone` / `camera`	`true` for each one fitted
`joints_offsets`	Measured per servo with `find_soft_offsets.py` (each servo sits a few degrees off)

Sources: the code on the duck (Open Duck Supercharged v2.5 runtime: `feet_contacts.py`, `antennas.py`, `eyes.py`, `projector.py`), the official assembly guide's pin table, and this build's own wiring decisions.